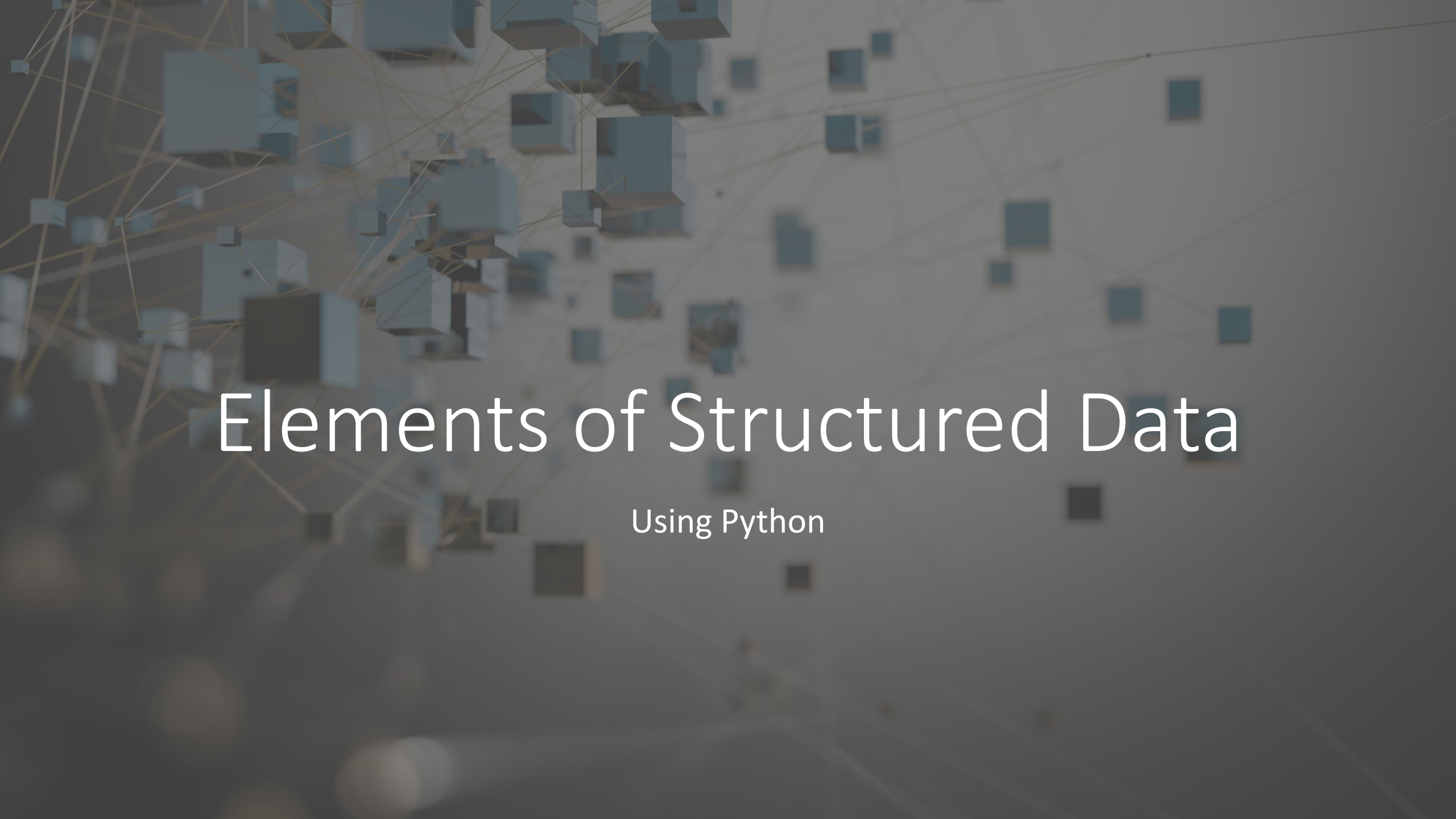


The background is a dark, muted grey-blue. It is populated with numerous small, three-dimensional blue cubes of varying sizes. These cubes are interconnected by a dense web of thin, golden-yellow lines, creating a complex, network-like structure that resembles a data graph or a molecular model. The overall aesthetic is modern and technological.

Elements of Structured Data

Using Python

Objectives

- Discuss the elements of structured data;
- Understand why data types needs to be identified for data analysis
- Define Rectangular data and how it is used in Python.
- Demonstrate manipulation of rectangular data through Google Colab.

Sources of data

- Data comes from many sources: sensor measurements, events, text, images, and videos.
- The Internet of Things (IoT) is spewing out streams of information. Much of this data is unstructured:
 - images are a collection of pixels, with each pixel containing RGB (red, green, blue) color information.
 - Texts are sequences of words and nonword characters, often organized by sections, subsections, and so on.

Sources of data

- A major challenge of data science is to harness this torrent of raw data into actionable information.
- To apply the statistical concepts, unstructured raw data must be processed and manipulated into a structured form.
- *One of the commonest forms of structured data is a table with rows and columns—as data might emerge from a relational database or be collected for a study.*

Two (2) Basic Types of Structured Data

Numerical-

- Numeric data comes in two forms:
 - *continuous*, such as wind speed or time duration, and
 - *discrete*, such as the count of the occurrence of an event.

Categorical-

- Categorical data takes only a fixed set of values, such as a type of TV screen (plasma, LCD, LED, etc.) or a state name (Alabama, Alaska, etc.).

Special Case of Categorical Data

- *Binary data* is an important special case of categorical data that takes on only one of two values, such as 0/1, yes/no, or true/false.
- Another useful type of categorical data is *ordinal data* in which the categories are ordered; an example of this is a numerical rating (1, 2, 3, 4, or 5).

Why do we bother with a taxonomy of data types?

- It turns out that for the purposes of data analysis and predictive modeling, the data type is important to help determine the type of visual display, data analysis, or statistical model.
- More important, the data type for a variable determines how software will handle computations for that variable.

Key Terms for Data Types

Key Terms for Data Types

Numeric

Data that are expressed on a numeric scale.

Continuous

Data that can take on any value in an interval. (*Synonyms*: interval, float, numeric)

Discrete

Data that can take on only integer values, such as counts. (*Synonyms*: integer, count)

Categorical

Data that can take on only a specific set of values representing a set of possible categories. (*Synonyms*: enums, enumerated, factors, nominal)

Binary

A special case of categorical data with just two categories of values, e.g., 0/1, true/false. (*Synonyms*: dichotomous, logical, indicator, boolean)

Ordinal

Categorical data that has an explicit ordering. (*Synonym*: ordered factor)

Rectangular Data

- The typical frame of reference for an analysis in data science is a *rectangular data object*, like a spreadsheet or database table.
- *Rectangular data* is the general term for a two-dimensional matrix with rows indicating records (cases) and columns indicating features (variables); *data frame* is the specific format in R and Python.
- The data doesn't always start in this form: *unstructured data (e.g., text) must be processed and manipulated so that it can be represented as a set of features in the rectangular data.*
- Data in relational databases must be extracted and put into a single table for most data analysis and modeling tasks.

Key Terms for Rectangular Data

Key Terms for Rectangular Data

Data frame

Rectangular data (like a spreadsheet) is the basic data structure for statistical and machine learning models.

Feature

A column within a table is commonly referred to as a *feature*.

Synonyms

attribute, input, predictor, variable

Outcome

Many data science projects involve predicting an *outcome*—often a yes/no outcome (in **Table 1-1**, it is “auction was competitive or not”). The *features* are sometimes used to predict the *outcome* in an experiment or a study.

Synonyms

dependent variable, response, target, output

Records

A row within a table is commonly referred to as a *record*.

Synonyms

case, example, instance, observation, pattern, sample

Table with description of data

- Mix of measured or counted data (duration, price)
- Special form of categorical variable (binary- yes/no or 0/1)
- Indicator variable (showing whether an auction is competitive or not)-
 - This indicator happens to be an outcome variable, when the scenario is to predict whether an auction is competitive or not.

Table 1-1. A typical data frame format

Category	currency	sellerRating	Duration	endDay	ClosePrice	OpenPrice	Competitive?
Music/Movie/Game	US	3249	5	Mon	0.01	0.01	0
Music/Movie/Game	US	3249	5	Mon	0.01	0.01	0
Automotive	US	3115	7	Tue	0.01	0.01	0
Automotive	US	3115	7	Tue	0.01	0.01	0
Automotive	US	3115	7	Tue	0.01	0.01	0
Automotive	US	3115	7	Tue	0.01	0.01	0
Automotive	US	3115	7	Tue	0.01	0.01	1
Automotive	US	3115	7	Tue	0.01	0.01	1

Data Frames and Indexes

- Traditional database tables have one or more columns designated as an index, essentially a row number.
- This can vastly improve the efficiency of certain database queries.
- In Python, with the *Pandas library*, the basic rectangular data structure is a *DataFrame* object.
- By default, an automatic integer index is created for a *DataFrame* based on the order of the rows.

What is indexing?

- Indexing is the way to get an unordered table into an order that will maximize the query's efficiency while searching.

Estimates of Location

- Variables with measured or count data might have thousands of distinct values.
- A basic step in exploring your data is getting a “typical value” for each feature (variable): an estimate of where most of the data is located (i.e., its central tendency).

Key Terms for Estimates of Location

Mean

- The sum of all values divided by the number of values.
- Synonym: average

Median

- The value such that one-half of the data lies above and below.
- Synonym: 50th percentile

Percentile

- The value such that P percent of the data lies below.
- Synonym: quantile

Key Terms for Estimates of Location

Outlier

- A data value that is very different from most of the data.
- Synonym: extreme value

Robust

- Not sensitive to extreme values.
- Synonym: resistant

Mean

- The most basic estimate of location is the mean, or average value.
- *The mean is the sum of all values divided by the number of values.*
- Consider the following set of numbers: {3 5 1 2}.
- The mean is $(3 + 5 + 1 + 2) / 4 = 11 / 4 = 2.75$.
- You will encounter the symbol $\bar{x}$ (pronounced “x-bar”) being used to represent the mean of a sample from a population.

Median

- *The median is the middle number on a sorted list of the data.*
- If there is an even number of data values, the middle value is one that is not actually in the data set, but rather the average of the two values that divide the sorted data into upper and lower halves.
- Compared to the mean, which uses all observations, the median depends only on the values in the center of the sorted data.
- While this might seem to be a disadvantage, since the mean is much more sensitive to the data, there are many instances in which the median is a better metric for location.

Outliers

- The median is referred to as a robust estimate of location since it is not influenced by outliers (extreme cases) that could skew the results.
- *An outlier is any value that is very distant from the other values in a data set.*
- Being an outlier in itself does not make a data value invalid or erroneous.
- Still, outliers are often the result of data errors such as mixing data of different units (kilometers versus meters) or bad readings from a sensor.
- When outliers are the result of bad data, the mean will result in a poor estimate of location, while the median will still be valid.

Getting the Measures of Central Tendency using Google Colab and Python

- There are two ways to create a data:
 1. Defining dictionary containing data and converting it into DataFrame.
 2. Import a .csv file containing data.

To perform data manipulation and data analysis, a software library called Pandas will be used for this discussion.

What is Pandas?

- Pandas is a Python library for data analysis.
- Started by Wes McKinney in 2008 out of a need for a powerful and flexible quantitative analysis tool, pandas has grown into one of the most popular Python libraries.
- Pandas is built on top of two core Python libraries—[matplotlib](#) for data visualization and [NumPy](#) for mathematical operations.
- Pandas acts as a wrapper over these libraries, allowing you to access many of matplotlib's and NumPy's methods with less code.

Why use Pandas?

- Pandas allows us to analyze big data and make conclusions based on statistical theories.
- Pandas can clean messy data sets and make them readable and relevant.
- Relevant data is very important in data science.

What Can Pandas Do?

- Pandas gives you answers about the data. Like:
- Is there a correlation between two or more columns?
- What is average value?
- Max value?
- Min value?
- Pandas are also able to delete rows that are not relevant, or contains wrong values, like empty or NULL values.
- This is called *cleaning* the data.

Pandas as pd

- Pandas is usually imported under the pd alias.
- **alias:** In Python alias are an alternate name for referring to the same thing.
- Create an alias with the as keyword while importing:
- `import pandas as pd`
- Now the Pandas package can be referred to as *pd* instead of pandas.

Sample Code: Defining Dictionary

```
#import pandas as package
import pandas as pd

#define a dictionary containing data
data={'Name': ['Juan dela Cruz', 'Mark Reyes','Chris Lopez','Angela Delgado', 'Sam Medina'],
      'Age':[20, 21, 19, 20, 22],
      'Course': ['CS','IT', 'EE','ECE','COE'],
      'Year Level':[2,3,1,2,4]
    }
```

```
[2] #import pandas as package
import pandas as pd

#define a dictionary containing data
data={'Name': ['Juan dela Cruz', 'Mark Reyes','Chris Lopez','Angela Delgado', 'Sam Medina'],
      'Age':[20, 21, 19, 20, 22],
      'Course': ['CS','IT', 'EE','ECE','COE'],
      'Year Level':[2,3,1,2,4]
    }

print (data)
```

```
{'Name': ['Juan dela Cruz', 'Mark Reyes', 'Chris Lopez', 'Angela Delgado', 'Sam Medina'], 'Age': [20, 21, 19, 20, 22], 'Course': ['CS', 'IT', 'EE', 'ECE', 'COE']}
```

Sample Code: Defining Dictionary

- After defining the dictionary containing the data, convert it to Data Frame to be able to display in tabular format.

✓
0s

```
#Convert the dictionary into DataFrame  
ddf=pd.DataFrame(data)
```

```
#Convert the dictionary into DataFrame  
ddf=pd.DataFrame(data)  
print(ddf)
```

	Name	Age	Course	Year	Level
0	Juan dela Cruz	20	CS		2
1	Mark Reyes	21	IT		3
2	Chris Lopez	19	EE		1
3	Angela Delgado	20	ECE		2
4	Sam Medina	22	COE		4

Sample Code: Defining Dictionary

- Once you have created the data frame, you can now start performing exploratory data analysis (getting the measures of central tendency, measures of spread, etc).

Getting the mean

```
▶ #getting the mean
statemean=df[['Population']].mean()
print (statemean)
```

```
Population    7690760.625
dtype: float64
```

```
[ ] #getting the median
statemean=df[['Population']].median()
print (statemean)
```

```
Population    4163416.5
dtype: float64
```

```
▶ #getting the mode
statemode=df[['Murder rate']].mode()
print (statemode)
```

```
Murder rate
0          5.6
```

Practice

1. Create a dictionary based on the given data below.

	A	B	C	D	E
1	Name	Subject	Grade	Number of days present	Number of days absent
2	Juan dela Cruz	CCE 101/L	89	3	7
3	Mark Reyes	CCE 101/L	86	1	4
4	Chris Lopez	CCE 101/L	88	2	5
5	Angela Lopez	CCE 101/L	92	2	0
6	Sam Medina	CCE 101/L	88	1	2
7					
8					

2. Then, create and generate data frame.

3. Get the mean for grades, number of days present, and number of days absent.